

Task 1: List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

Error Message	Usual Hardware / Configuration Issue
CMOS checksum error – Defaults loaded	Corrupted or lost BIOS settings, often caused by a dead/weak CMOS battery or a sudden power loss
Display Adapter Not Found" or a 1 Long, 2 Short Beep Code	The system is failing its initial video check. This indicates that the dedicated graphics card (GPU) is either not receiving sufficient power from the PSU PCIe cables, is not fully snapped into the motherboard's PCIe x16 slot, or its onboard hardware has suffered a failure.
No boot device found	BIOS can't detect a bootable hard drive/SSD — could mean a disconnected drive cable, wrong boot order in BIOS, or a failed/corrupted drive
Memory size decreased / Memory test failed	A RAM module failed, not properly seated, or has been removed/changed since the last boot
CPU fan error / CPU fan not detected	The CPU cooling fan isn't spinning or isn't connected to the correct motherboard header, risking overheating if ignored

Task 2: Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

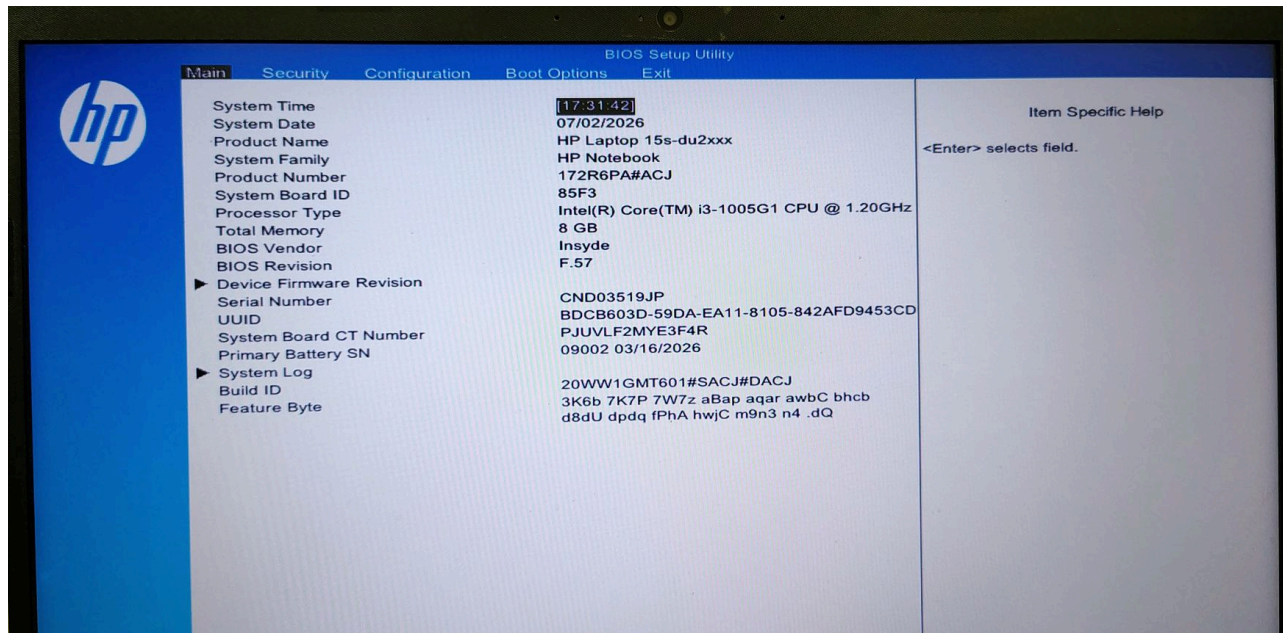
On AMI (American megatrends) BIOS, 3 short beeps indicate a “Base 64K memory failure” — During the initial Power-On Self-Test (POST), the BIOS tests the very first 64 kilobytes of the system's Random Access Memory (RAM), which is critical for initializing basic system instructions. A 3-beep failure means the motherboard cannot read or write to this foundational block of memory, halting the boot process before the display can even be initialized (hence the blank screen).

Two troubleshooting steps:

1. Reseat the RAM module — power off, unplug, open the case, remove the RAM stick(s), gently clean the gold contacts with a dry cloth or eraser, and firmly reinsert them until the clips click into place. A loose connection is one of the most common causes of this exact beep code.

2. Test with a single RAM stick at a time (if you have multiple modules) — remove all but one stick and try booting, then swap to test each module and each slot individually. This isolates whether the fault is a specific bad stick or a bad slot on the motherboard.

Task 3: Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



Board ID: 85F3

BIOS Vendor: Insyde

Processor Support: Integrated Intel Core i3-1005G1 CPU



Based on Board ID and code on the motherboard (LS-H324P) - the board manufacturer is likely COMPAL Electronics

Task 4: You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.

It means:

The computer's BIOS settings (like the date, time, and boot order) got corrupted or lost. So instead of using your saved settings, the BIOS loads its factory defaults and shows this warning message.

It is usually triggered by:

- A dead or dying CMOS battery (most common cause — this battery keeps the settings and clock powered when the PC is off)
- A sudden power loss or improper shutdown while BIOS settings were being written
- Rarely, a corrupted BIOS chip itself

How to resolve it in a real system:

1. Replace the CMOS battery — it's a small coin-shaped battery (usually a CR2032) on the motherboard. This fixes the problem most of the time.
2. Go into BIOS and set things up again — after replacing the battery, enter BIOS setup, fix the date/time and boot order, then save and exit.

If it keeps happening even with a new battery, the problem is likely bigger than just the battery — could be a motherboard fault that may need a technician.